

Algorithms Problem Solving

Level 1

Hire Driver Case1

The background of the slide features a large whiteboard with a complex network diagram drawn on it. The diagram consists of numerous interconnected nodes and lines, resembling a web or a circuit. In the foreground, the silhouettes of several people are visible, appearing to be in a meeting or collaborative work environment. They are positioned around the whiteboard, looking at the diagram.

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Definition

- ◆ This program reads a person's age and whether they have a driving license, then determines whether they are eligible to be hired.
- ◆ The program uses a structure, functions, a Boolean value, and a logical AND condition.

Explanation

- ◆ The struct `stInfo` groups the user's related information:
 - `age` stores the person's age.
 - `hasDrivingLicense` stores whether the person has a driving license as a bool value.
 - `readInfo()` reads the user's information and returns an `stInfo` object.
 - `isValidInput()` checks both conditions:
 - The age must be greater than or equal to 21.
 - The person must have a driving license.
- ◆ The `&&` operator means both conditions must be true.
- ◆ `printResult()` calls `isValidInput()` and prints `Hired` if the result is true; otherwise, it prints `Rejected`.
- ◆ In `main()`, the functions are combined:
- ◆ `printResult(readInfo());`

Steps in main()

1. Create the stInfo structure.
2. Read the user's age.
3. Read whether the user has a driving license.
4. Return the collected information.
5. Check if age ≥ 21 and hasDrivingLicense is true.
6. Return true or false.
7. Print Hired if the result is true; otherwise print Rejected.

💡 Example

◆ If the user enters:

Age: 25

Driving license: 1

Both conditions are true.

➡ Result

Hired

◆ If the user enters:

Age: 25

Driving license: 0

The driving license condition is false.

➡ Result

Rejected

Note

- ◆ The bool value `hasDrivingLicense` is read as 1 for true and 0 for false..



Comparison

Hired

- ✓ Age is 21 or greater.
- ✓ Has a driving license.
- ✓ Both conditions must be true because of &&.

Rejected

- ✓ Age is below 21, or
- ✓ Does not have a driving license.
- ✓ At least one condition is false.

✦✦ Summary

- ✓ A structure can group related information into one object.
- ✓ The && operator requires both hiring conditions to be true.
- ✓ Functions can work together by passing returned values from one function to another.

Thank you 😊